

**END TO END DAX SCENARIO
QUESTIONS
FOR EXPERIENCED POWER
BI ROLE**

1. Data Modeling & Context (Foundation)

1. You have Sales, Customer, Product, Date tables. How do you design relationships for correct DAX evaluation?
2. What happens internally when you use ALL vs ALLSELECTED vs REMOVEFILTERS?
3. A measure works in a table visual but gives wrong results in a card. Why?
4. How does row context convert to filter context? Explain using SUMX.
5. When do you prefer calculated columns vs measures in a real project?
6. How do you handle bi-directional relationships and their impact on DAX?

2. Time Intelligence (Very Important)

7. Create YTD, QTD, MTD sales using DAX.
8. Sales data exists only by month. How do you calculate YoY growth?
9. What happens if the date table is not marked? How does it affect SAMEPERIODLASTYEAR?
10. How do you calculate rolling 3 / 6 / 12 months sales?
11. Explain DATESINPERIOD vs DATEADD with a scenario.

12. Fiscal year starts in April. How do you handle FY-based calculations?

3. Ranking, Top N & Others (Real-World Scenarios)

13. Show Top 5 customers by sales with a slicer to change N dynamically.

14. How do you create Top N + Others in a single visual?

15. Why does Top N filter behave differently from RANKX-based logic?

16. Create a Top / Bottom toggle using a slicer.

17. Rank products within each category, not globally.

18. Handle ranking when sales values are tied.

4. Advanced Filter Context Scenarios

19. Calculate % contribution of each customer to total sales.

20. Show running total without a date column.

21. How do you calculate previous row value in DAX?

22. Sales should ignore Region slicer but respect Date slicer. How?

23. Explain a case where ALL breaks your visual.

24. How do you debug filter context issues in complex measures?

5. Business KPIs & Conditional Logic

- 25. Categorize customers as High / Medium / Low value dynamically.
- 26. Create a KPI that shows Green if growth > 10%, Yellow if between 0–10%, Red if negative.
- 27. Switch measures dynamically using a measure selector.
- 28. Show different calculations for Retail vs Corporate customers.
- 29. Handle division by zero in KPI calculations.

6. Performance Optimization (Must for Experienced Role)

- 30. Why is SUM better than SUMX in most cases?
- 31. How do variables (VAR) improve DAX performance?
- 32. Explain a real scenario where a DAX measure caused slow visuals.
- 33. How do you reduce cardinality issues in DAX?
- 34. Difference between FILTER inside CALCULATE vs WHERE in SQL.

7. Real Project Debugging Scenarios

- 35. A report works in Dev but fails in Prod. How do you debug DAX?

36. Totals are wrong in a matrix visual. How do you fix it?

37. SAMEPERIODLASTYEAR returns blank. What are the possible reasons?

38. Measures behave differently after publishing to Service.

8. Security & Advanced Scenarios

39. Implement dynamic Row-Level Security (RLS) using USERPRINCIPALNAME().

40. Restrict users to see only their own sales but full totals.

41. Can RLS impact DAX results? Explain with an example.

9. Composite & Multi-Fact Scenarios

42. Combine Budget vs Actual vs Forecast in one visual.

43. Compare current month sales vs same month last year vs average.

44. Create a measure that behaves differently at Customer level, Region level, and Grand total.

45. Handle many-to-many relationships using DAX.

10. Interview Twist Questions (Advanced)

46. Why does CALCULATE() work even without filters?

- 47. Explain context transition without using the term itself.
- 48. Can a measure change the data model? Why or why not?
- 49. Why do we avoid calculated columns for aggregations?
- 50. Explain a real business problem you solved using DAX.